

WILLIAM RUDMAN

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EDUCATION

Brown University | Providence, RI

2020 – 2025

Ph.D. in Computer Science

Research Areas: Representation Learning & Mechanistic Interpretability

Advisor: Carsten Eickhoff

King's College London | London, England

2018 – 2019

M.Sc. Complex Systems Modelling, Pass with Distinction

Complex Systems Modelling Prize: Best Master's Thesis

Thesis: *Frustration Solitaire with Stochastic Resetting*

Advisor: Pierpaolo Vivo

University of Aberdeen | Aberdeen, Scotland

2014 – 2018

Joint M.A. in Mathematics and Sociology, First Class Honours

Thesis: *Representation Theory of Finite Groups*

Advisor: Jean-Baptiste Gramain

ACADEMIC APPOINTMENTS

The University of Texas at Austin | Austin, TX

August 2025 – Present

Postdoctoral Research Fellow

Computational Linguistics & the NSF-Simons AI Institute for Cosmic Origins

Advisors: Kyle Mahowald & Junyi Jessy Li

PUBLICATIONS

- [1] **William Rudman**, Abhishek Divekar, Kanishk Jain, Sebastian Joseph, Stella S. R. Offner, Matthew Lease, Kyle Mahowald, Greg Durrett, and Junyi Jessy Li. *VESTA: Visual Exploration with Statistical Tool Agents*. Neural Information Processing Systems: NeurIPS 2026.
- [2] Florian Rottach*, **William Rudman***, Bastian Rieck, Harrison Scells, and Carsten Eickhoff. *From Topology to Retrieval: Decoding Embedding Spaces with Unified Signatures*. Neural Information Processing Systems: NeurIPS 2026.
- [3] Niclas Lietzow, Danielle Bitterman, Carsten Eickhoff, **William Rudman***, and Michal Golovanevsky*. *Vision-Default, Prior-Override: Causal Mechanisms of Perception-Knowledge Conflict in Vision-Language Models*. Findings of the Association for Computational Linguistics: EMNLP 2026.
- [4] Kaijie Mo, Thomas Yang, Chantal Shaib, Qing Yao, **William Rudman**, Ramez Kouzy, Kanishka Misra, Byron C. Wallace, and Junyi Jessy Li. *What's in a Name? Morphological Shortcuts by LLMs in Pharmacology*. Empirical Methods in Natural Language Processing: EMNLP 2026.
- [5] Jonathan Nemitz, Carsten Eickhoff, Junyi Jessy Li, Kyle Mahowald, Michal Golovanevsky*, and **William Rudman***. *When to Call an Apple Red: Humans Follow Introspective Rules, VLMs Don't*. Conference on Language Modeling: COLM 2026.
- [6] **William Rudman***, Michal Golovanevsky*, Dana Arad, Yonatan Belinkov, Ritambhara Singh, Carsten Eickhoff, and Kyle Mahowald. *Mechanisms of Prompt-Induced Hallucination in Vision-Language Models*. Association for Computational Linguistics: ACL 2026.
- [7] **William Rudman***, Michal Golovanevsky*, Amir Bar, Michael Lepori, Ritambhara Singh, and Carsten Eickhoff. *Pixels Versus Priors: Controlling Knowledge Priors in Vision-Language Models through Visual Counterfactuals*. Empirical Methods in Natural Language Processing: EMNLP 2025.
- [8] **William Rudman***, Michal Golovanevsky*, Amir Bar, Vedant Palit, Ritambhara Singh, Carsten Eickhoff, and Yann LeCun. *Forgotten Polygons: Multimodal Large Language Models are Shape-Blind*. Findings of the Association for Computational Linguistics: ACL 2025.

- [9] **William Rudman***, Michal Golovanevsky*, Vedant Palit, Ritambhara Singh, and Carsten Eickhoff. *What Do VLMs NOTICE? A Mechanistic Interpretability Pipeline for Noise-free Text-Image Corruption and Evaluation*. Nations of the Americas Chapter of the Association for Computational Linguistics: NAACL 2025.
- [10] **William Rudman** and Carsten Eickhoff. *Stable Anisotropic Regularization*. International Conference on Learning Representations: ICLR 2024.
- [11] **William Rudman**, Catherine Chen, and Carsten Eickhoff. *Outlier Dimensions Contain Task-Specific Knowledge*. Empirical Methods in Natural Language Processing: EMNLP 2023.
- [12] William Jurayj, **William Rudman**, and Carsten Eickhoff. *Garden Path Traversal in GPT-2*. Proceedings of BlackboxNLP 2022.
- [13] **William Rudman**, Nate Gillman, Taylor Rayne, and Carsten Eickhoff. *IsoScore: Measuring the Uniformity of Embedding Space Utilization*. Findings of the Association for Computational Linguistics: ACL 2022.
- [14] **William Rudman**, Peter Corcoran, and Kimberly Elliot. *Prescription Drug Monitoring Programs and Opioid Death Rates*. Journal of the American Medical Association: JAMA 2017.

WORKSHOPS

- [1] Florentin Beck, **William Rudman**, and Carsten Eickhoff. *TRIM: Achieving Extreme Sparsity with Targeted Row-wise Iterative Metric-driven Pruning*. COLM 2026 Workshop on Efficient Reasoning.
- [2] Frauke Andersen, **William Rudman**, Ruochen Zhang, and Carsten Eickhoff. *Accelerated Path Patching with Task-Specific Pruning*. BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks 2025.
- [3] Isaac Sears, Augusto Garcia-Agundez, George Zerveas, **William Rudman**, Laura Mercurio, Corey E. Ventetuolo, Adeel Abbasi, and Carsten Eickhoff. *Leveraging Unlabeled Electroencephalographic Data to Predict Neurological Recovery for Comatose Patients Following Cardiac Arrest*. Computing in Cardiology: CinC 2023.
- [4] **William Rudman**, Jack Merullo, and Carsten Eickhoff. *ACQuA: Anomaly Classification with Quasi-Attractors*. Computing in Cardiology: CinC 2022.

PRE-PRINTS

- [1] Florian Rottach, Sebastian Schieferdecker, **William Rudman**, Randall Balestriero, and Carsten Eickhoff. *Mol-JEPA: A multimodal Joint Embedding Predictive Architecture for Molecules*. Under Review 2026.
- [2] Josh Barua, **William Rudman**, and Kyle Mahowald. *Vision-Language Models Integrate Conflicting Evidence by Reliability*. Under Review 2026.
- [3] Okan S. Coskun, Florian Rottach, Carsten Eickhoff, and **William Rudman**. *Isotropy Cliffs: The Geometric Signature of Decision-Making in Large Language Models*. Under Review 2026.
- [4] Yating Wu, **William Rudman**, Venkata S. Govindarajan, Alexandros G. Dimakis, and Junyi Jessy Li. *Multimodal QUD: Inquisitive Questions from Scientific Figures*. Under Review 2026.
- [5] Oliver McLaughlin, Daniel Shubin, Carsten Eickhoff, Ritambhara Singh, **William Rudman***, and Michal Golovanevsky*. *Is There Knowledge Left to Extract? Evidence of Fragility in Medically Fine-Tuned Vision-Language Models*. Under Review 2026.

WORK EXPERIENCE

PayPal San Jose, CA Machine Learning Research Scientist Intern	June 2024 – September 2024
Lincoln College of Technology Chicago, IL Adjunct Instructor: Technical Mathematics	January 2020 – September 2020
Rosalind Franklin University of Medicine and Science North Chicago, IL Public Health Research Intern	June 2017 – September 2017

GRANTS, HONORS, AND AWARDS

- ITSERR Research Fellow: National Research Council of Italy

April 2025

- T32 NIH Predoctoral Biological Data Science Fellowship September 2020 – May 2022
- King’s College London Complex Systems Modeling Prize: Best Master’s Thesis December 2019
- King’s College London International Master’s Scholar September 2018

PROFESSIONAL ACTIVITIES

Guest Lectures & Invited Talks

- University of Pisa: Guest Lecture for *Mechanistic Interpretability of LLMs* April 2026
- UT-Austin: Guest Lecture for *Discourse Structure in LLMs* November 2025
- UT-Austin: Computational Linguistics Department May 2025
- MIT: Geometric Data Processing Group April 2025
- National Research Council of Italy March 2025
- University of Hawaii at Manoa: AI Precision Health Institute December 2023
- New England NLP: UMass Lowell April 2023

Reviewing: EMNLP (2023-2026), ACL (2024-2026), COLM (2026), ICML (2026), BlackboxNLP (2026), EACL (2025-2026), NAACL (2025), ICLR (2025), NeurIPS (2025), SIGGRAPH (2025), OpenMind (2025).

SKILLS

Programming Languages: Python, PyTorch, Pandas, Git, NumPy, DeepSpeed, PySpark, SQL

Technical Knowledge: Natural Language Processing, Large Language Models, Vision-Language Models, Agentic-AI, Mechanistic Interpretability, Machine Learning, Statistics